

2018 Ph H1 Q16

Section: Electricity

Topic: Current, PD, Power, Resistance

Question Summary

A $120\ \Omega$ resistor dissipates a power of $4.8\ \text{W}$. What is the current through the resistor?

Worked Solution

Use the power relationship $P = I^2R$.

Rearrange for current: $I = \sqrt{P/R}$.

Substitute values: $I = \sqrt{4.8 / 120} = \sqrt{0.040} = 0.20\ \text{A}$.

Final Answer: C ($0.20\ \text{A}$)

Revision Tips

- Choose the power formula that matches the given data: $P = VI$, $P = I^2R$, or $P = V^2/R$.
- When you know P and R , use $I = \sqrt{P/R}$.
- Keep units consistent: W , Ω , A .
- Use appropriate significant figures; here, 2 s.f. is suitable ($0.20\ \text{A}$).